

Acids and Bases

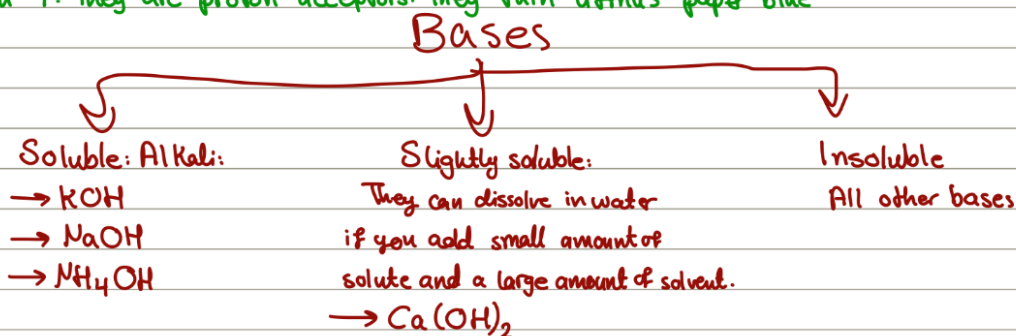
Acids: Non-metal compounds in water. Contain hydrogen ions, which is what makes them acidic. They have a pH lower than 7. They are proton (H^+) donors. They turn litmus paper red.

* Hydrochloric acid: HCl

* Sulfuric acid: H_2SO_4

* Nitric acid: HNO_3

Bases: Metal oxides and hydroxides. Contain hydroxide ions, which is what makes them basic. They have a pH higher than 7. They are proton acceptors. They turn litmus paper blue



	Acid	Base	Neutral
Litmus	Red	Blue	→ When red, stays red. When blue, stays blue
Methyl orange	Red	Yellow	Red
Thymolphthalein	Colorless	Blue	Colorless
Phenolphthalein	Colorless	Pink	Colorless

Universal indicator colors:

1-2: Red 3-4: Orange 5: Yellow 6: Yellowish-green 7: Green 8-9: Greenish-Blue
10-12: Blue 13-14: Violet

* A dilute strong acid is still stronger than a concentrated weak acid.

Concentrated, dilute: The amount of substance

Strong, weak: The H^+ concentration in aqueous solution

* In strong acids, all molecules dissociate completely (Become ions). In weak acids, not all molecules dissociate completely.

Hydrochloric acid forms chloride salts

Sulfuric acid forms sulfate salts

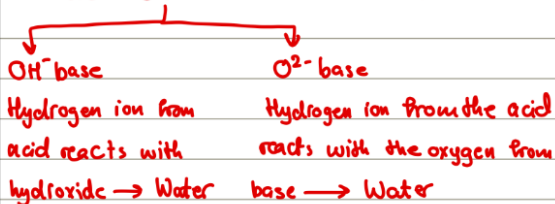
Nitric acid forms nitrate salts

Acid Reactions:

1- Acid + Metal → Salt + Hydrogen

* the metal drives the hydrogen out of the acid, and takes its place. (It displaces the hydrogen)

2- Acid + Base → Salt + Water



3- Acid + Metal carbonate → Salt + Water + Carbon dioxide

* Neutralization reaction: A reaction with acid that gives out water as well as salt Metal carbonate
→ Bases

Base reactions:

1- Base + acid $\rightarrow$ Salt + Water

2- Base + Ammonium salt $\rightarrow$ Salt + Water + Ammonia gas

→ KOH
→ NaOH
→ Ca(OH)_2

Oxides: Compounds that contain oxygen and another element. Metal oxides: Bases, Non-metal oxides: Acids

① Basic oxides:

- Magnesium oxide: MgO : Forms when magnesium reacts with oxygen

Observations: 1- Burns with a brilliant white flame

2- Forms white ash: MgO

- Iron (III) oxide: Fe_2O_3 : Forms when iron(III) reacts with oxygen

Observations: 1- Glows bright orange

2- Throws out a shower of sparks

3- Forms a black solid: Fe_2O_3

- Copper(II) oxide: CuO : Forms when copper reacts with oxygen.

Observation: Surface turns black

② Acidic oxides:

- Carbon dioxide: Forms when carbon reacts with oxygen: CO_2

Observation: Glows Bright red.

- Sulfur dioxide: Forms when sulfur reacts with oxygen: SO_2

Observation: Burns with a blue flame.

- Phosphorus pentoxide: Forms when phosphorus reacts with oxygen: P_4O_{10}

Observation: Bursts into flame without heating.

③ Amphoteric oxides: Reacts with both acids and bases

- Aluminum oxide: Al_2O_3

- Zinc oxide: ZnO

- Chromium oxide: Cr_2O_3

4- Neutral oxides: Do not react with acids nor bases.

- Carbon monoxide: CO

- Dinitrogen oxide: N_2O

Solvent: The liquid in which the solute dissolves in.

Solute: The substance which dissolves in solvent

Solution: The mixture formed when a solute dissolves in a solvent

Saturated solution: A solution with the maximum concentration of solute dissolved in solvent

Solubility of a substance: The amount of the substance that dissolves in 100g of water at a specific temperature

Filtrate: The liquid or solution that has passed through filter paper

Residue: The substance that remains after evaporation, distillation or filtration.

* When you heat a solution, kinetic energy will increase, which results in more collisions between particles, so more solute dissolves in solvent. A saturated solution would become unsaturated solution.

When do you know that all the acid has been used up completely in:

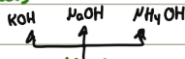
1- Metals: - Bubbling stops

- Precipitate formation (unreacted metal)

2- Metal carbonate: - Bubbling stops

- Precipitate formation (unreacted metal)

3- Insoluble base: Precipitate formation



4- Alkali: We use titration: Add phenolphthalein to an alkali, it will turn pink. Then add the acid using a burette to the alkali. Keep adding acid until the color of the solution turns from pink to colorless, which indicates neutralization. Which means all the base have been neutralized by the acid. Repeat the experiment by adding the same amount of alkali and acid without adding phenolphthalein, as it is considered an impurity.

How to separate salts from solution?

1- If the salt was insoluble, then filter the solution. The residue left is the salt, the water will pass through as the filtrate.

2- If the salt was insoluble, then we use crystallization. Heat the solution to evaporate some of the water, so it becomes more concentrated; eventually, the solution becomes saturated. When you cool it down, crystals start to form. Remove the crystals then by filtering, and rinse them with distilled water and dry the crystals.